AI BASED CYLINDER COUNTING & INSPECTION SYSTEM

AI-Powered Offline Cylinder Counting, Cap Detection & Inspection System

Prepared By ComSys IT

INTRODUCTION

WHAT IT IS ?

The AI-Based Cylinder Counting, Cap Detection & Multi-Angle Inspection System is an offline solution that automates the cylinder unloading process using industrial cameras and Artificial Intelligence (AI). The system counts cylinders, verifies the presence of safety caps, captures images from multiple angles, and generates digital reports.

WHY IT IS NEEDED ?

- Eliminate manual counting errors.
- Improve unloading speed and operational efficiency.
- Ensure every cylinder has a safety cap.
- Maintain digital records with image evidence.
- Enhance safety, accuracy, and process transparency.

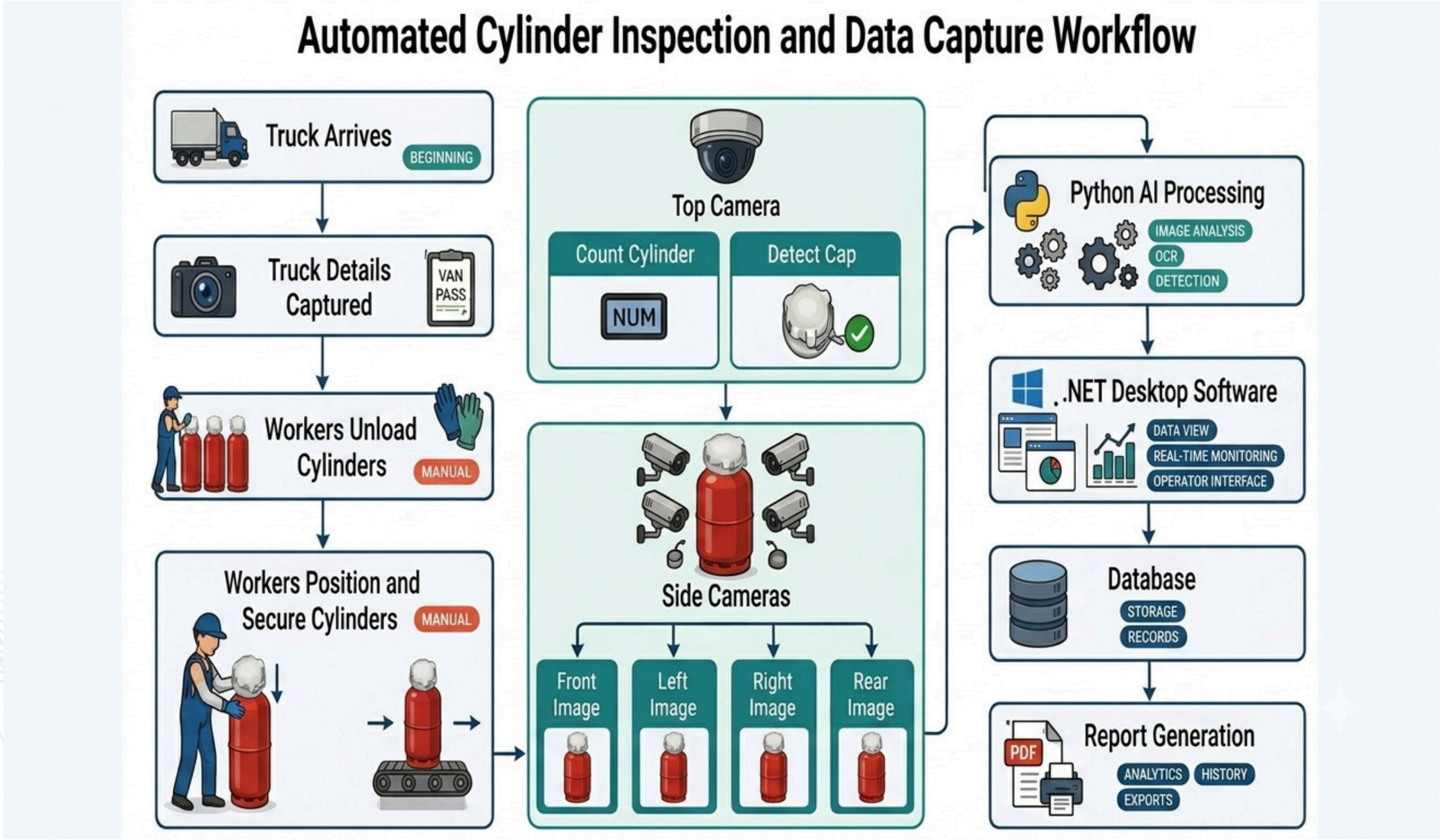


PROPOSED SOLUTION

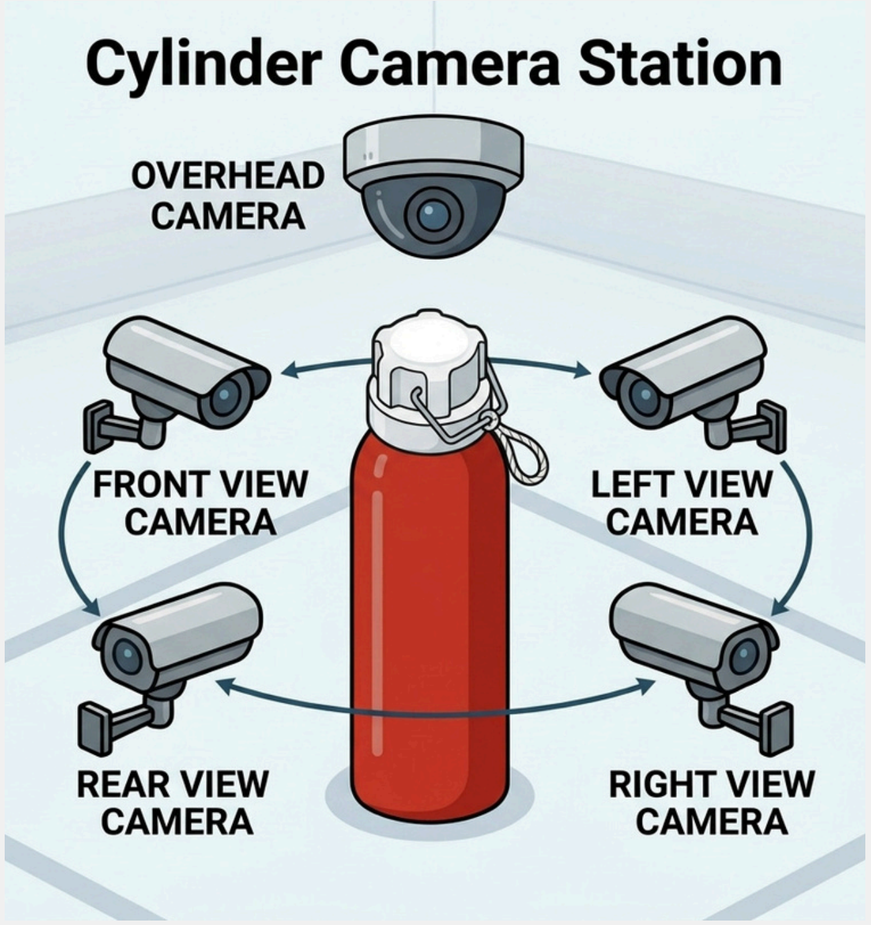
WHEN A TRUCK ENTERS THE UNLOADING BAY,

1. TRUCK ARRIVES AT UNLOADING BAY.
2. APPLICATION RECORDS TRUCK INFORMATION.
3. WORKERS UNLOAD CYLINDERS.
4. CYLINDERS MOVE ONE BY ONE ON CONVEYOR.
5. AI DETECTS EVERY CYLINDER.
6. AI CHECKS WHETHER CAP IS PRESENT.
7. MULTIPLE CAMERAS CAPTURE CYLINDER IMAGES.
8. COUNT IS UPDATED AUTOMATICALLY.
9. FINAL INSPECTION REPORT IS GENERATED.

COMPLETE WORKFLOW



CAMERA PLACEMENT



Camera 1 (Top)
Purpose

- Cylinder Detection
- Counting
- Cap Detection

Camera 2 (Left side image)

Camera 3 (Right side image)

Camera 4/5 Based on site requirement (Front/Rear image)

Purpose

- Complete cylinder documentation
- Image verification
- Damage inspection (Future Enhancement)

REPORT GENERATION

After the unloading process is completed, the system automatically generates a detailed inspection report. The report includes truck details, total cylinder count, cap detection results, captured images, date and time, and inspection status. This digital report can be stored, searched, or exported for future reference and auditing. Given below is just a demo and not an actual report.

Summary by Unloading Bay & Truck ID

Truck Number	Arrival Time	Bay ID	Worker Count	Cylinders Counted	Cap Pass Status	AI Verification Status
20000	09:08:30	1	2	442		Verified
20002	09:09:10	2	1	450		Verified (2 Missing Caps)
20003	06:09:30	3	1	421		(2 Missing Caps)
Total Anomalies Flagged			3	442	8	

ADVANTAGES

- ✔ Fully Offline
- ✔ High Accuracy
- ✔ AI Based Image Ana
- ✔ Faster than Manual Counting
- ✔ Automatic Report Generation
- ✔ Digital Records
- ✔ Image Proof
- ✔ Detect Missing Caps
- ✔ Scalable
- ✔ Low Human Error

THE PROPOSED AI-BASED SYSTEM AUTOMATES CYLINDER COUNTING AND INSPECTION, REDUCING MANUAL EFFORT AND MINIMIZING HUMAN ERRORS. IT IMPROVES OPERATIONAL EFFICIENCY BY PROVIDING ACCURATE CYLINDER COUNTS, AUTOMATIC CAP VERIFICATION, DIGITAL IMAGE RECORDS, AND INSTANT REPORT GENERATION. AS THE SOLUTION OPERATES COMPLETELY OFFLINE, IT ALSO ENSURES DATA SECURITY AND RELIABLE PERFORMANCE WITHIN THE PLANT ENVIRONMENT.

IMPLEMENTATION ROADMAP & TIMELINE

A structured, 4-phase approach to ensure a smooth and successful deployment.

[PHASE 1: HARDWARE INSTALLATION]
Mount HD cameras at the inspection bay. Install the necessary cabling and networking infrastructure to support camera connectivity and data transmission.
Estimated Time: 1 - 2 Weeks

[PHASE 2: DATA COLLECTION & AI MODEL TRAINING]
Capture and label 1,500-2,000 images per truck type based on location. Train and validate the AI model for counting and cap detection.
Estimated Time: 3 - 4 Weeks

[PHASE 3: SERVER SETUP & DEPLOYMENT]
Install the AI processing APPLICATION on-site. Integrate the system with existing plant operations and reporting systems.
Estimated Time: 1 - 2 Weeks

[PHASE 4: TESTING & GO-LIVE]
Conduct pilot runs, perform iterative refinements, and execute full-scale production deployment.
Estimated Time: 2 - 3 Weeks

THANK YOU

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